

AI Engineer — SOC Agent Platform

Main Interview Loop · Panther Labs

• 4 Interviews via Zoom • Engineering · IC5 • San Francisco, CA

This guide walks you through each interview in the main loop for the AI Engineer — SOC Agent Platform role. For each interview you'll find what we're evaluating, the specific signals we assess, and targeted advice on how to prepare.

ABOUT THE ROLE

Panther is building the AI SOC platform that modern security teams need. You'll work directly with our Founder & CTO to build AI-powered SOC analyst solutions — autonomous agents for alert triage, interactive chat, detection code generation, and text-to-search — deeply integrated with Panther's ingestion pipeline. You'll also engage directly with customers to validate agent effectiveness in real SOC environments.

We're looking for engineers who combine **strong agentic development skills**, **security domain knowledge**, and a **startup ownership mindset**.

THE MAIN INTERVIEW LOOP

WHAT THIS INTERVIEW IS ABOUT

A high-level design conversation to confirm you have the technical acumen to build **production-grade AI agents**. You'll design an AI-powered SOC alert triage agent, walk through architecture decisions, embedding and retrieval strategy, deterministic vs. LLM-driven logic, and how the system would operate in production. **Clarity of reasoning matters more than perfect syntax.**

SIGNALS WE'RE EVALUATING

Agent Architecture Depth

Layered agentic loops, tool design, composition, evaluation — not just LLM wrappers

RAG & Retrieval Design

Embeddings, vector DBs, multi-layered pipelines, token budget management

Prompt Engineering at Scale

Composable, maintainable prompt systems across multiple conversation types

Security / SOC Fluency

What's safe to automate vs. what requires human judgment in a SOC context

Production Engineering Maturity

Evaluation, monitoring, drift detection, failure modes — not prototype thinking

AI Security Awareness

Prompt injection, adversarial inputs, defense-in-depth in a security product

Product & Customer Thinking

Measurable outcomes, customer trust, phased autonomy over time

HOW TO PREPARE

- **Know your agent architecture fundamentals.** Be ready to walk through an agentic loop end-to-end: context pre-loading, tool calling, deterministic vs. LLM steps, async inference, structured output, post-inference processing.
- **Know your RAG stack.** Discuss multi-layered retrieval, token budget constraints, and handle "50K matching events in a 200K token context window."
- **Think prompt engineering at scale.** How do you avoid 12 monolithic prompt strings that drift independently? Composable modules, shared bases, testing.
- **Know AI security blindspots.** Prompt injection is real when log events are attacker-controlled. Structural defenses + RBAC at the tool execution boundary.
- **Think in production metrics.** Offline eval sets, online metrics (analyst override rate, classification distribution), regression detection, feedback loops.
- **Brush up on:** RAG, LLM tool calling, MCP (Model Context Protocol), agentic loop design, prompt composition, confidence scoring, SOC alert triage workflows.

WHAT THIS INTERVIEW IS ABOUT

Evaluates your ability to handle **ambiguous technical problems** and design scalable, well-reasoned systems. You'll walk through system architecture decisions — processes, storage/databases, networking, APIs — and discuss tradeoffs, prioritization, and cross-team collaboration. **Not a coding exercise.**

SIGNALS WE'RE EVALUATING

Systems Thinking

End-to-end design with clear data abstractions, API boundaries, storage strategies

Technical Reasoning

Articulating tradeoffs — not just "it depends" but what and why

Problem-Solving Under Ambiguity

Good clarifying questions, defining success criteria, systematic approach

Failure Mode Awareness

Proactively thinking about what can go wrong and how to detect/mitigate it

Collaboration & Communication

Explaining technical decisions to non-technical stakeholders

Strategic Fit

Good judgment, initiative, alignment with how we think about security and AI

HOW TO PREPARE

- **Pick 1–2 complex projects to discuss deeply.** Ideally AI, automation, or agentic systems. Key architectural tradeoffs, what you'd do differently.
- **Practice designing an agentic system from scratch.** Data flow, storage decisions, API design, concurrency, latency, and failure handling.
- **Think in tradeoffs.** Explain why you made each decision and what the alternatives were. Latency vs. cost. Consistency vs. availability.
- **Come with good questions.** Curiosity about our technical challenges and architectural decision-making signals strong alignment.

WHAT THIS INTERVIEW IS ABOUT

A deep-dive into a **recent, complex technical project you've led end-to-end** from concept to launch. You choose the project. No coding or technical execution — a structured conversation about a real initiative you drove.

SIGNALS WE'RE EVALUATING

Project Scope

Meaningful initiative requiring real technical leadership and cross-functional coordination

Technical Ownership

Genuine end-to-end accountability, deep understanding of the full stack

Business Impact

Connecting engineering decisions to business and customer outcomes

Measurement of Success

Concrete metrics — not just "it shipped" but what changed and by how much

Problem Solving

Biggest technical challenges, how you solved them, tradeoffs made

Collaboration & Influence

Working with PMs, designers, engineers, business stakeholders

CHOOSING YOUR PROJECT

Completed in the last 18 months

At least 3 months in duration

Cross-functional (PM, design, other engineering teams, business stakeholders)

Within the domain of this role (AI/ML engineering, backend systems, security tooling)

You had genuine end-to-end ownership — not just one contributor among many

COME PREPARED TO DISCUSS

- Your specific scope and responsibilities; project duration, team size, and functions involved
- Technology stack and key architectural decisions with tradeoffs
- The business problem, how you measured success, and the biggest technical challenge
- The outcome, quantified if possible — and what you'd do differently

WHAT THIS INTERVIEW IS ABOUT

Panther's culture is built on three core values: **Create Customer Love**, **Be an Owner**, and **Take Care of the Team**. This interview focuses on how you've demonstrated these values through real, specific professional examples. Domain skills are assessed elsewhere — this round is about **grit, autonomy, and the attributes that make competence actionable**.

SIGNALS WE'RE EVALUATING

 Create Customer Love

Direct customer engagement; building things that solve real problems over technically interesting ones

 Take Care of the Team

Supporting teammates, sharing knowledge, prioritizing team success over personal credit

 Startup Mindset

Energized by building from scratch; comfortable with high ownership and fast movement

 Be an Owner

End-to-end accountability, proactively fixing problems, founder-level responsibility

 Grit & Ambiguity

Making progress in undefined environments; handling failure, pivots, unclear requirements

 Self-Awareness

Honest identification of strengths and growth areas; learning from mistakes

HOW TO PREPARE

- **Prepare 4–6 specific stories** from your career demonstrating Panther's core values. Specific, recent, showing your individual impact.
- **For each story:** Situation → your specific action → outcome → what you'd do differently.
- **Think about times you've:** gone above and beyond for a customer; owned a problem without being asked; stepped up for a struggling teammate; made a hard call under ambiguity.
- **Be honest and specific.** Authentic self-reflection outperforms polished stories. Failures discussed candidly are often more compelling than success stories.

Key Themes Across All Interviews

Production Thinking Over Prototypes

Real security teams rely on what you build. Demonstrate reliability, evaluation, monitoring, and failure mode thinking.

Customer Obsession

Connect technical decisions to customer outcomes. Why does this architecture choice matter? Because it reduces triage time for analysts.

Clarity of Communication

How you explain your thinking matters as much as what you think. Structure your answers and explain your reasoning.

Security Awareness

This is a security product. Show you understand the unique risks of automating decisions in high-stakes environments.

Startup Ownership

We move fast with a small team. End-to-end accountability, decisions without perfect information, care about the outcome.

Depth Over Breadth

We're evaluating whether you've actually built these systems — not whether you can recite definitions. Go deep on real experience.